# APM52009 – Machine Learning for Scientific Computing and Numerical Analysis

Week 6: Introduction to Parametric PDEs and Reduced Basis Methods

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**Abstract**

Many problems in science and engineering require solving the same PDE repeatedly for different parameter values, leading to prohibitively large computational costs when high-fidelity discretizations are used. Reduced Basis (RB) methods address this many-query challenge by exploiting the fact that parametric PDE solutions often lie close to a low-dimensional manifold. In this chapter, we introduce RB methods for linear parametric elliptic PDEs, focusing on the offline/online decomposition, Galerkin projection onto reduced spaces, and practical construction of reduced bases with the SVD. We show how model reduction can drastically reduce computational complexity while preserving accuracy, and we provide theoretical insight—based on compactness, analyticity, and Kolmogorov $N$-widths—explaining why such reductions are possible for smooth elliptic problems.

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# 1 Why Reduced Order Models?

Modern numerical methods can approximate PDE solutions with high accuracy. Yet this comes at a price: after discretization, even simple PDEs often yield linear systems with millions of unknowns. Solving one system is feasible, but repeating it thousands of times quickly becomes impractical.

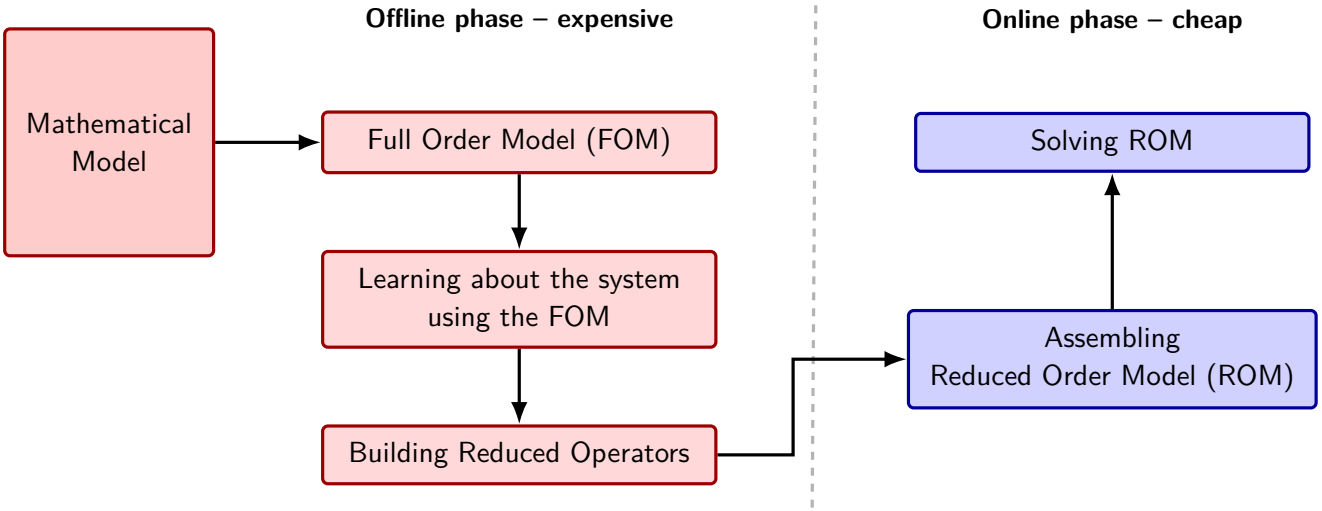
This situation arises routinely in *many-query* settings, where the same PDE must be solved repeatedly for different values of a parameter vector $\boldsymbol{\mu}$. In these contexts, the computational bottleneck is no longer accuracy for a single solve, but efficiency across repeated queries.

**Key observation** Although each high-fidelity solution lives in a very high-dimensional space, the set of all solutions obtained by varying the parameters often exhibits much lower intrinsic complexity. As the parameter $\boldsymbol{\mu}$ varies, the solution $u(\boldsymbol{\mu})$ typically explores only a small region of the full discretization space. Geometrically, the solutions concentrate near a low-dimensional *solution manifold* embedded in a high-dimensional ambient space.

**What reduced order models do** Reduced Order Models (ROMs) exploit this structure by replacing the original high-dimensional problem—the Full Order Model (FOM)—with a much smaller one. The idea is to:

1. compute a small number of representative high-fidelity solutions of the FOM called *snapshots*;
2. extract from them a low-dimensional space that captures the dominant variability;
3. project the governing equations onto this reduced space.

Once this reduced space has been constructed, new parameter queries can be solved extremely fast, often in real time.




**Offline vs. online** A central feature of ROMs is the separation between:

- an *offline phase*, where expensive computations are performed once;
- an *online phase*, where new solutions are computed at negligible cost.

This separation is what makes ROMs attractive in many-query regimes: high accuracy is retained, while computational cost becomes essentially independent of the original discretization size.

**Scope of this chapter** We focus on RB methods, a specific type of ROM, for linear parametric elliptic PDEs. RB methods construct a low-dimensional subspace from a set of high-fidelity solutions (snapshots) of the FOM and use this space to define a reduced Galerkin system. We explain how reduced spaces are built via snapshot data and the SVD, how reduced Galerkin systems are derived, and why—under suitable regularity assumptions—these methods achieve rapid convergence. The theoretical results presented later provide a mathematical justification for why low-dimensional linear approximations are effective in this setting.

## 2 Parametric PDEs

**Motivation** Many problems in science and engineering require the repeated solution of a PDE for different values of a set of parameters. Let $\boldsymbol{\mu} \in \mathcal{P} \subset \mathbb{R}^{d_P}$ denote a vector of parameters (e.g., material coefficients, geometric features, boundary data, source terms), where $\mathcal{P}$ is a compact parameter set.

After spatial discretization, each parameter value $\boldsymbol{\mu}$ leads to a linear system

$$A_h(\boldsymbol{\mu})U_h(\boldsymbol{\mu}) = F_h(\boldsymbol{\mu}), \tag{2.1}$$

where $A_h(\boldsymbol{\mu}) \in \mathbb{R}^{N_h \times N_h}$ represents a discretized differential operator, $F_h(\boldsymbol{\mu}) \in \mathbb{R}^{N_h}$ a discretized source term, and $U_h(\boldsymbol{\mu})$ the vector of degrees of freedom.

In many applications, this system must be solved for *many* values of $\boldsymbol{\mu}$:

- in optimization or inverse problems, where a PDE is solved at every iteration;
- in nonlinear problems, where a linearized system is solved repeatedly;
- in time-dependent or multi-query settings, where solutions are required in real time.

When $N_h$ is large, repeatedly solving this high-dimensional system quickly becomes computationally prohibitive. This observation is the starting point of *reduced order modeling*.

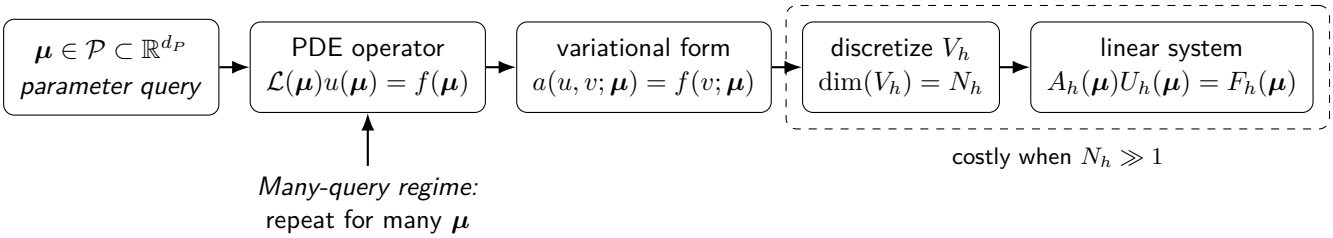



Figure 1: Typical parametric PDE workflow—in many-query settings, repeatedly assembling/solving the high-dimensional system becomes the bottleneck.

**Mathematical setup** Let $D \subset \mathbb{R}^d$ be a bounded domain and $V = V(D)$ a Hilbert space (e.g., $H_0^1(D)$), with dual space $V'$.

**Problem 2.1** (Strong formulation)**.** For every parameter value $\boldsymbol{\mu} \in \mathcal{P}$, we consider a second-order elliptic operator $\mathcal{L}(\boldsymbol{\mu}) : V \to V'$, together with right-hand side $f(\boldsymbol{\mu}) \in V'$ such that

$$\mathcal{L}(\boldsymbol{\mu})u(\boldsymbol{\mu}) = f(\boldsymbol{\mu}) \quad \text{in } V'.$$

The goal is to determine the solution map

$$\boldsymbol{\mu} \in \mathcal{P} \longmapsto u(\boldsymbol{\mu}) \in V,$$

and to approximate it efficiently for many parameter values.

From this perspective, the parametric PDE defines a map $\boldsymbol{\mu} \mapsto u(\boldsymbol{\mu})$, whose efficient approximation is the central object of interest in reduced order modeling.

We denote by $(\cdot,\cdot)_V$ the inner product defined over $V$ and by $\|\cdot\|_V$ the induced norm. Let $\langle \cdot, \cdot \rangle$ denote the duality pairing between $V'$ and $V$.

Introducing the bilinear and linear forms $a(\cdot,\cdot;\boldsymbol{\mu}) : V \times V \to \mathbb{R}$ and $f(\cdot;\boldsymbol{\mu}) : V \to \mathbb{R}$

$$a(u, v; \boldsymbol{\mu}) := \langle \mathcal{L}(\boldsymbol{\mu})u, v \rangle, \qquad f(v; \boldsymbol{\mu}) := \langle f(\boldsymbol{\mu}), v \rangle,$$

the strong formulation can be rewritten equivalently as the following variational problem.

**Problem 2.2** (Variational formulation)**.** Given $\boldsymbol{\mu} \in \mathcal{P}$, find $u(\boldsymbol{\mu}) \in V$ such that

$$a(u(\boldsymbol{\mu}), v; \boldsymbol{\mu}) = f(v; \boldsymbol{\mu}) \quad \forall v \in V.$$

If, for each $\boldsymbol{\mu} \in \mathcal{P}$, $a(\cdot, \cdot; \boldsymbol{\mu})$ is $C(\boldsymbol{\mu})$-continuous and $\alpha(\boldsymbol{\mu})$-coercive and $f(\cdot; \boldsymbol{\mu})$ is continuous, then there exists a unique solution $u(\boldsymbol{\mu})$ for each $\boldsymbol{\mu} \in \mathcal{P}$, and

$$\|u(\boldsymbol{\mu})\|_V \leq \frac{1}{\alpha(\boldsymbol{\mu})} \|f(\cdot; \boldsymbol{\mu})\|_{V'} \quad \forall \boldsymbol{\mu} \in \mathcal{P}. \tag{2.2}$$

We discretize $V$ by $\{V_h\}_{h>0}$, a sequence of nested, dense finite-dimensional subspaces of $V$ as $h \to 0$.

**Problem 2.3** (Galerkin approximation)**.** Given $\boldsymbol{\mu} \in \mathcal{P}$, find $u_h(\boldsymbol{\mu}) \in V_h$ such that

$$a(u_h(\boldsymbol{\mu}), v_h; \boldsymbol{\mu}) = f(v_h; \boldsymbol{\mu}) \quad \forall v_h \in V_h.$$

The bilinear form $a(\cdot, \cdot; \boldsymbol{\mu})$ is continuous and $\alpha(\boldsymbol{\mu})$-coercive on $V_h \times V_h$, hence the problem admits a unique solution for each $\boldsymbol{\mu} \in \mathcal{P}$. Moreover, the Galerkin approximation satisfies the quasi-optimal error estimate

$$\|u(\boldsymbol{\mu}) - u_h(\boldsymbol{\mu})\|_V \leq \frac{C(\boldsymbol{\mu})}{\alpha(\boldsymbol{\mu})} \inf_{v_h \in V_h} \|u(\boldsymbol{\mu}) - v_h\|_V \qquad \forall \boldsymbol{\mu} \in \mathcal{P}.$$

Let $\{\phi_i\}_{i=1}^{N_h}$ be a basis of $V_h$, and let $U_h(\boldsymbol{\mu}) \in \mathbb{R}^{N_h}$ denote the coefficient vector of $u_h$, i.e.,

$$u_h(\boldsymbol{\mu}) = \sum_{j=1}^{N_h} (U_h(\boldsymbol{\mu}))_j \, \phi_j.$$

Define $A_h(\boldsymbol{\mu}) \in \mathbb{R}^{N_h \times N_h}$ and $F_h(\boldsymbol{\mu}) \in \mathbb{R}^{N_h}$ via

$$(A_h(\boldsymbol{\mu}))_{ij} := a(\phi_j, \phi_i; \boldsymbol{\mu}), \qquad (F_h(\boldsymbol{\mu}))_i := f(\phi_i; \boldsymbol{\mu}).$$

The Galerkin approximation leads to the following linear system.

**Problem 2.4** (Galerkin linear system)**.** Given $\boldsymbol{\mu} \in \mathcal{P}$, find $U_h(\boldsymbol{\mu}) \in \mathbb{R}^{N_h}$ such that

$$A_h(\boldsymbol{\mu}) U_h(\boldsymbol{\mu}) = F_h(\boldsymbol{\mu}).$$

**Key observation** While $U_h(\boldsymbol{\mu}) \in \mathbb{R}^{N_h}$ is high-dimensional, the set of all discrete solutions

$$\mathcal{M}_h := \{U_h(\boldsymbol{\mu}) \ : \ \boldsymbol{\mu} \in \mathcal{P}\} \subset \mathbb{R}^{N_h},$$

often exhibits a much lower intrinsic dimension. Reduced order methods aim to exploit this structure to replace the high-dimensional problem by a much smaller one, while retaining accuracy and stability.

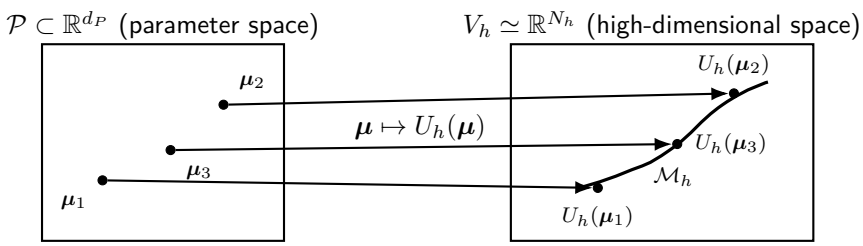



Figure 2: The parameter-to-solution map. Even though $U_h(\boldsymbol{\mu}) \in \mathbb{R}^{N_h}$, the set of solutions often concentrates near a low-dimensional manifold, motivating reduced-order models.

Reduced order methods aim to construct a low-dimensional space $V_N \subset V_h$, with $\dim(V_N) = N \ll N_h$, that accurately captures the solution manifold $\mathcal{M}_h$. The original high-dimensional problem is then projected onto $V_N$, yielding a reduced system that can be solved efficiently for new parameter values. The following sections describe the construction of such spaces and the mechanisms ensuring both efficiency and stability.

# 3 Reduced basis methods

**Goal and guiding idea** For each parameter value $\boldsymbol{\mu} \in \mathcal{P}$, the high-fidelity approximation $u_h(\boldsymbol{\mu}) \in V_h$ is obtained by solving a linear system of size $N_h$. RB methods address the costs of many-query solves by exploiting the observation that the set of solutions $\mathcal{M}_h$ often has *low intrinsic dimension*: although $V_h$ is large, the solution map $\boldsymbol{\mu} \mapsto u_h(\boldsymbol{\mu})$ typically explores only a small portion of $V_h$.

## 3.1 From high-fidelity to reduced Galerkin systems

**High-fidelity space** Let $\{\phi_i\}_{i=1}^{N_h}$ be a basis of the high-fidelity space $V_h$. Any discrete solution is represented by a coefficient vector

$$U_h(\boldsymbol{\mu}) \in \mathbb{R}^{N_h}, \qquad u_h(\boldsymbol{\mu}) = \sum_{j=1}^{N_h} (U_h(\boldsymbol{\mu}))_j \, \phi_j.$$

For each parameter value $\boldsymbol{\mu} \in \mathcal{P}$, the coefficients satisfy Problem 2.4.

**Reduced space** RB methods introduce a low-dimensional subspace $V_N \subset V_h$ of dimension $N \ll N_h$, with basis functions $\{\psi_i\}_{i=1}^{N}$. Each reduced basis function can be expanded in the high-fidelity basis:

$$\psi_i = \sum_{j=1}^{N_h} (W_i)_j \, \phi_j, \qquad i = 1, \dots, N,$$

where $W_i \in \mathbb{R}^{N_h}$ is the coefficient vector of $\psi_i$ in the high-fidelity basis. Collecting these vectors as columns defines the matrix

$$W = \Big[ W_1 \,|\, \cdots \,|\, W_N \Big] \in \mathbb{R}^{N_h \times N}.$$

The reduced solution is sought as a linear combination of the reduced basis functions, equivalently in terms of the high-fidelity coefficients:

$$U_h(\boldsymbol{\mu}) \approx W \, U_N(\boldsymbol{\mu}), \qquad U_N(\boldsymbol{\mu}) \in \mathbb{R}^N.$$

**Projection** Substituting the reduced ansatz into the full system yields the overdetermined system

$$A_h(\boldsymbol{\mu}) \, W \, U_N(\boldsymbol{\mu}) = F_h(\boldsymbol{\mu}),$$

consisting of $N_h$ equations for $N$ unknowns. The reduced solution is defined by enforcing that the residual be orthogonal to the reduced space:

$$W^\top \big( A_h(\boldsymbol{\mu}) \, W \, U_N(\boldsymbol{\mu}) - F_h(\boldsymbol{\mu}) \big) = 0.$$

**Reduced Galerkin system** Defining

$$A_N(\boldsymbol{\mu}) := W^\top A_h(\boldsymbol{\mu}) W \in \mathbb{R}^{N \times N}, \qquad F_N(\boldsymbol{\mu}) := W^\top F_h(\boldsymbol{\mu}) \in \mathbb{R}^N,$$

the reduced system can be written compactly as the square system of size $N \times N$.

**Problem 3.1** (Reduced Galerkin linear system)**.** Given $\boldsymbol{\mu} \in \mathcal{P}$, find the solution $U_N(\boldsymbol{\mu}) \in \mathbb{R}^N$ of

$$W^\top A_h(\boldsymbol{\mu}) W \, U_N(\boldsymbol{\mu}) = W^\top F_h(\boldsymbol{\mu}).$$

**Remark 3.2.** At the coefficient level, the columns $W_i$ of the matrix $W \in \mathbb{R}^{N_h \times N}$ are the coefficient vectors of the reduced basis functions in the high-fidelity basis $\{\phi_j\}$. The reduced solution is then expressed as $U_h(\boldsymbol{\mu}) \approx W \, U_N(\boldsymbol{\mu})$, and the reduced system corresponds to enforcing orthogonality of the residual in $\mathbb{R}^{N_h}$ with respect to the columns of $W$. At the function level, each reduced $\psi_i$ reads

$$\psi_i = \sum_{j=1}^{N_h} (W_i)_j \, \phi_j,$$

so that the reduced space is $V_N = \operatorname{span}\{\psi_1, \dots, \psi_N\} \subset V_h$. Thus, the reduced system corresponds to projecting the full-order operator onto the subspace $V_N$ in the original function space.

Visually, we have the following *offline/online* decomposition. In SciML terminology, the offline phase corresponds to training, while the online phase would be evaluating the model (i.e., a forward pass).

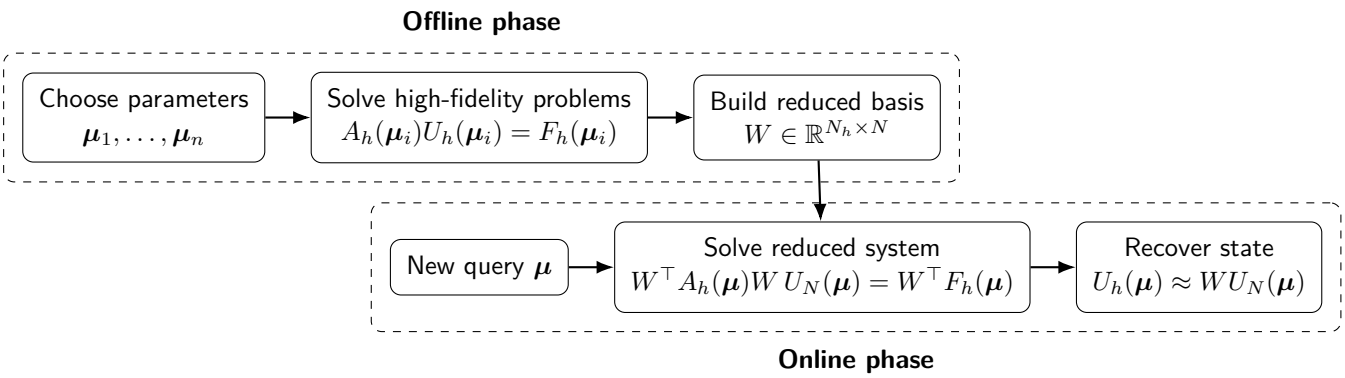



Figure 3: Offline/online decomposition in reduced basis methods. The reduced basis matrix $W$ is computed once (offline) and reused for many parameter queries (online).

**Remark 3.3** (What RB *does not* solve automatically)**.** The reduced dimension $N \ll N_h$ guarantees a small linear system *once* $A_N(\boldsymbol{\mu})$ and $F_N(\boldsymbol{\mu})$ are available. However, assembling $A_N(\boldsymbol{\mu}) = W^\top A_h(\boldsymbol{\mu})W$ can still be expensive if one forms $A_h(\boldsymbol{\mu})$ explicitly for every new $\boldsymbol{\mu}$. Efficient online assembly typically requires additional structure, e.g., affine parameter dependence.

**Example 3.4** (Affine parameter dependence)**.** A common situation in RB methods is that the bilinear form admits an *affine decomposition* with respect to the parameter $\boldsymbol{\mu}$:

$$a(u, v; \boldsymbol{\mu}) = \sum_{q=1}^{Q} C_q(\boldsymbol{\mu})\, a_q(u, v),$$

where the coefficient functions $C_q : \mathcal{P} \to \mathbb{R}$ depend only on the parameter and the bilinear forms $a_q(\cdot, \cdot)$ are parameter-independent. At the matrix level, this yields

$$A_h(\boldsymbol{\mu}) = \sum_{q=1}^{Q} C_q(\boldsymbol{\mu})\, A_h^{(q)}, \qquad (A_h^{(q)})_{ij} = a_q(\phi_j, \phi_i).$$

Consequently, the reduced matrix can be assembled as

$$A_N(\boldsymbol{\mu}) = W^\top A_h(\boldsymbol{\mu})W = \sum_{q=1}^{Q} C_q(\boldsymbol{\mu})\, A_N^{(q)}, \qquad A_N^{(q)} := W^\top A_h^{(q)} W,$$

where the matrices $A_N^{(q)} \in \mathbb{R}^{N \times N}$ are precomputed offline once. The online assembly cost is therefore $\mathcal{O}(QN^2)$, independent of $N_h$.

**Takeaways**

- The RB strategy is to construct a low-dimensional subspace $V_N \subset V_h$ that approximates $\mathcal{M}_h$ well, and to replace the expensive $N_h$-dimensional solve by a cheap $N$-dimensional solve.
- RB methods trade a (moderate) *offline* cost—building a reduced space from a few high-fidelity solutions—for a very fast *online* stage, where new queries $\boldsymbol{\mu}$ are solved in dimension $N \ll N_h$.

Now, the main goal in the offline stage is to construct the reduced basis functions $\psi_i$ (or equivalently the matrix $W$) that define the reduced space $V_N$.

## 3.2 Constructing a reduced basis

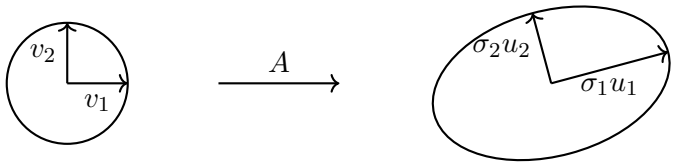
**What we need** We seek a low-dimensional space $V_N = \mathrm{span}\{\psi_1, \dots, \psi_N\} \subset V_h$ that approximates the snapshot set $\{u_h(\boldsymbol{\mu}_j)\}_{j=1}^n$ well. At the coefficient level, this becomes a *matrix approximation problem*: given snapshot vectors $U_h(\boldsymbol{\mu}_j) \in \mathbb{R}^{N_h}$, we want an orthonormal matrix $W \in \mathbb{R}^{N_h \times N}$ with $W^\top W = I_N$ such that the projections $WW^\top U_h(\boldsymbol{\mu}_j)$ minimize the reconstruction error ($WW^\top$ is a projection since $(WW^\top)^2 = WW^\top$).

**Remark 3.5.** Reduced basis construction is fundamentally a *compression* task: find the best $N$-dimensional subspace that captures the dominant variability of the snapshots.

**Singular value decomposition** Let $A \in \mathbb{R}^{m \times n}$ with $m \geq n$. The (reduced) SVD reads

$$A = U\Sigma V^\top \qquad \Longleftrightarrow \qquad AV = U\Sigma, \tag{3.1}$$

with orthogonal $U \in \mathbb{R}^{m \times n}$ and $V \in \mathbb{R}^{n \times n}$, and $\Sigma = \mathrm{diag}(\sigma_1, \dots, \sigma_n)$ with $\sigma_1 \geq \cdots \geq \sigma_n \geq 0$. Geometrically, $A$ maps the unit ball in $\mathbb{R}^n$ to an ellipsoid in $\mathbb{R}^m$; the vectors $v_j$ give the principal directions in the domain and the vectors $u_j$ give the principal directions in the image, with axis lengths $\sigma_j$.




$\{v_j\}$ right singular vectors in $\mathbb{R}^n$ $\qquad$ $\{u_j\}$ left singular vectors in $\mathbb{R}^m$

Figure 4: Geometric meaning of the SVD: $A$ maps the unit ball to an ellipsoid whose principal axes are $\sigma_j u_j$.

**Theorem 3.6** (Facts about the SVD)**.**

1. *Every $A \in \mathbb{R}^{m \times n}$ has a SVD $A = U\Sigma V^\top$; computing it costs $\mathcal{O}(mn^2 + n^3)$.*
2. *Singular values are unique; singular vectors are not unique.*
3. *The nonzero singular values of $A$ are the square roots of the nonzero eigenvalues of $A^\top A$ and $AA^\top$; the left (right) singular vectors are eigenvectors of $AA^\top$ ($A^\top A$).*
4. *The rank $r$ equals the number of nonzero singular values; $\mathrm{range}(A) = \mathrm{span}(u_1, \dots, u_r)$ and $\mathrm{null}(A) = \mathrm{span}(v_{r+1}, \dots, v_n)$.*
5. *$\|A\|_2 = \sigma_1$ and $\|A\|_F = \sqrt{\sum_{j=1}^r \sigma_j^2}$, with $\|A\|_2 \leq \|A\|_F$.*
6. *The rank-$N$ truncation $A_N = \sum_{i=1}^N \sigma_i u_i v_i^\top$ is the best rank-$N$ approximation of $A$:*
$$\|A - A_N\|_2 = \min_{\mathrm{rank}(B) \leq N} \|A - B\|_2 = \sigma_{N+1}, \quad \|A - A_N\|_F = \min_{\mathrm{rank}(B) \leq N} \|A - B\|_F = \sqrt{\sum_{j=N+1}^r \sigma_j^2}.$$
7. *The pseudo-inverse of $A$ is $A^\dagger = V\Sigma^\dagger U^\top \in \mathbb{R}^{n \times m}$ with $\Sigma^\dagger = \mathrm{diag}(\sigma_1^{-1}, \dots, \sigma_r^{-1}, 0, \dots, 0)$.*
8. *If $A$ has full rank $n$, then*
$$A^\dagger = V\Sigma^{-1}U^\top = (A^\top A)^{-1}A^\top, \qquad \|A^\dagger\|_2 = 1/\sigma_n, \qquad \mathrm{cond}(A) = \|A\|_2 \cdot \|A^\dagger\|_2 = \sigma_1/\sigma_n.$$
9. *If $A \in \mathbb{R}^{n \times n}$ is square, then $\prod_{j=1}^n \sigma_j = |\det(A)|$, and if it is also of full rank $n$, then*
$$A^\dagger = A^{-1}, \qquad \|A^{-1}\|_2 = 1/\sigma_n, \qquad \mathrm{cond}(A) = \|A\|_2 \cdot \|A^{-1}\|_2 = \sigma_1/\sigma_n.$$
10. *Normal equations $A^\top Ax = A^\top b$ for linear least squares may be solved via $x = (V\Sigma^{-1}U^\top)b$.*

**Remark 3.7.** The most important statement is the *optimality of rank truncation*: keeping the first $N$ singular directions captures as much snapshot energy as any $N$-dimensional subspace (item #6).

**Application to RB methods** Assume $N_h \geq n$, and let $\{\boldsymbol{\mu}_1, \ldots, \boldsymbol{\mu}_n\} \subset \mathcal{P}$ be a set of parameter snapshots. Define the *snapshot matrix* collecting the high-fidelity solutions:

$$S = \Big[U_h(\boldsymbol{\mu}_1) \,|\, \cdots \,|\, U_h(\boldsymbol{\mu}_n)\Big] \in \mathbb{R}^{N_h \times n}.$$

Compute the SVD of $S$:

$$S = U\Sigma V^\top, \quad U \in \mathbb{R}^{N_h \times N_h}, \quad \Sigma \in \mathbb{R}^{N_h \times n}, \quad V \in \mathbb{R}^{n \times n},$$

and define the reduced basis matrix $W \in \mathbb{R}^{N_h \times N}$ by taking the first $N$ left singular vectors:

$$W = \Big[U_1 \,|\, \cdots \,|\, U_N\Big], \qquad W^\top W = I_N,$$

The rank-$N$ approximation of the snapshot matrix is

$$S_N = \sum_{i=1}^{N} \sigma_i U_i V_i^\top = WW^\top S \in \mathbb{R}^{N_h \times n},$$

**Theorem 3.8** (Approximating snapshots with the SVD)**.** *Let $S_N = WW^\top S$. Then*

$$\|S - S_N\|_F^2 = \sum_{j=N+1}^{r} \sigma_j^2 = \min_{B^\top B = I_N} \sum_{j=1}^{n} \|U_h(\boldsymbol{\mu}_j) - BB^\top U_h(\boldsymbol{\mu}_j)\|_2^2.$$

Proof

The first identity follows directly from the SVD truncation. For the optimality statement, observe that

$$\min_{B^\top B = I_N} \|S - BB^\top S\|_F^2 \geq \min_{\operatorname{rank}(B) \leq N} \|S - B\|_F^2 = \|S - S_N\|_F^2,$$

and choosing $B = W$ achieves equality.

A common criterion is to retain a prescribed fraction of snapshot energy: choose $N$ such that

$$\frac{\sum_{j=1}^{N} \sigma_j^2}{\sum_{j=1}^{r} \sigma_j^2} \geq 1 - \varepsilon^2,$$

which guarantees $\|S - S_N\|_F \leq \varepsilon \|S\|_F$. We are now ready to list our RB method with SVD.

**Algorithm 1** Reduced-basis construction with the SVD

1: Compute $n$ high-fidelity solutions and assemble $S = [U_h(\boldsymbol{\mu}_1)| \cdots |U_h(\boldsymbol{\mu}_n)] \in \mathbb{R}^{N_h \times n}$ (*offline*).
2: Compute the SVD $S = U\Sigma V^\top$.
3: Set $W = [U_1| \cdots |U_N] \in \mathbb{R}^{N_h \times N}$, choosing $N$ via $\sum_{j=1}^{N} \sigma_j^2 / \sum_{j=1}^{r} \sigma_j^2 \geq 1 - \varepsilon^2$.
4: For a new parameter $\boldsymbol{\mu}$ (*online*), solve $W^\top A_h(\boldsymbol{\mu}) W\, U_N(\boldsymbol{\mu}) = W^\top F_h(\boldsymbol{\mu})$ and output $U_h(\boldsymbol{\mu}) \approx W U_N(\boldsymbol{\mu})$.

**Proper Orthogonal Decomposition (POD/SVD)** The reduced basis $W$ obtained from the SVD of the snapshot matrix is also called the *Proper Orthogonal Decomposition (POD)*. It is *optimal for the snapshots*, minimizing their average Euclidean reconstruction error (see Theorem 3.8). This does *not* ensure uniform accuracy over all $\boldsymbol{\mu} \in \mathcal{P}$, nor optimality in other norms unless adapted.

**Offline/online computational complexity**

| **Operation** | **Sparse & affine** $A_h$ | **Sparse** $A_h$ | **Dense** $A_h$ |
|---|---|---|---|
| Compute snapshots $S = [U_h(\boldsymbol{\mu}_1) \,|\, \cdots \,|\, U_h(\boldsymbol{\mu}_n)]$ | $\mathcal{O}(nN_h)$ | $\mathcal{O}(nN_h)$ | $\mathcal{O}(nN_h^3)$ |
| Compute SVD of $S \in \mathbb{R}^{N_h \times n}$ | $\mathcal{O}(N_h n^2 + n^3)$ | | |
| Assemble reduced $A_N(\boldsymbol{\mu})$ | $\mathcal{O}(QN^2)$ | $\mathcal{O}(N_h N^2)$ | $\mathcal{O}(N_h^2 N)$ |
| Solve reduced system $A_N(\boldsymbol{\mu})U_N(\boldsymbol{\mu}) = F_N(\boldsymbol{\mu})$ | $\mathcal{O}(N^3)$ | | |

### 3.3 Extensions

The POD/SVD basis derived from the snapshot matrix $S$ is optimal for reconstructing the *training snapshots* in the Euclidean norm. However, standard POD may be insufficient when one wants to:

(i) capture time-dependent dynamics (snapshots across time);
(ii) enforce physically meaningful norms (e.g., energy norms);
(iii) approximate a *continuous* parameter distribution rather than a finite training set;
(iv) guarantee uniform accuracy over $\mathcal{P}$ (worst-case control).

The following approaches address these limitations.

| **Extension** | **What changes compared to basic POD/SVD?** |
|---|---|
| Time-dependent | snapshots indexed by $(\boldsymbol{\mu}, t)$ and reduced ODE system |
| Weighted/energy POD | replace $\|\cdot\|_2$ by $\|\cdot\|_{X_h}$ (inner product matters) |
| Continuous mode | replace training sum by integral over $\mathcal{P}$ (sampling/weights matter) |
| Worst-case control | target $\sup_{\boldsymbol{\mu}\in\mathcal{P}}$ rather than average error |

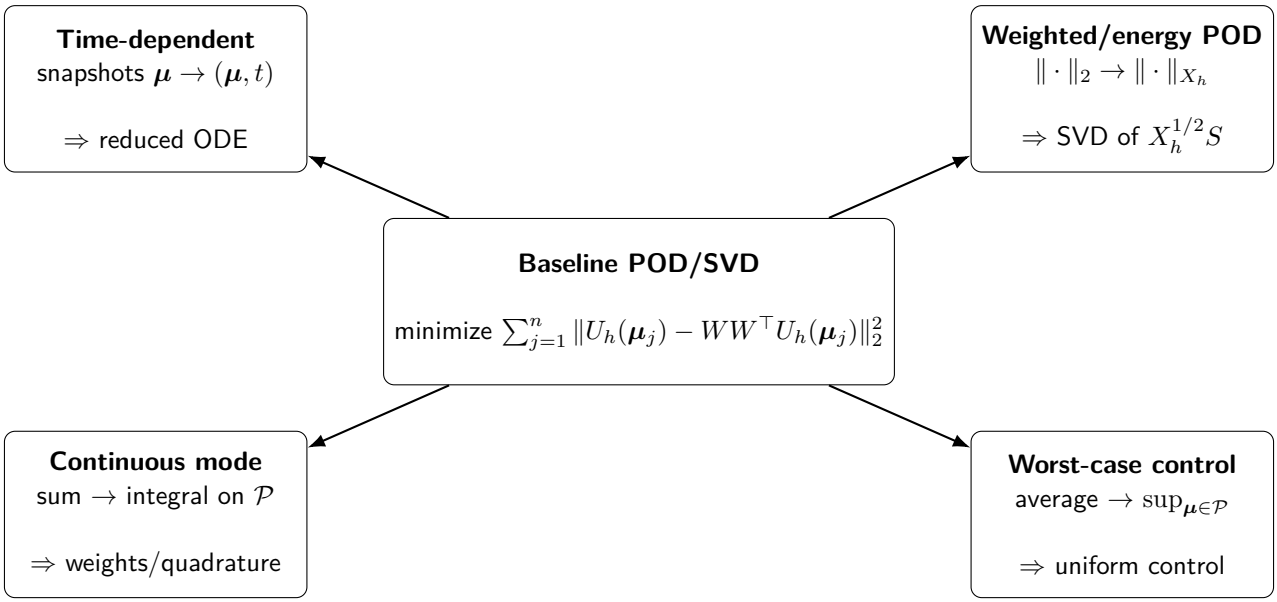



Figure 5: Extensions can be viewed as changing the baseline objective: add time, change the norm, change the measure on $\mathcal{P}$, or target worst-case accuracy.

**(i) Time-dependent problems: adding a temporal dimension** For time-dependent PDEs, snapshots are indexed not only by the parameter $\boldsymbol{\mu}$ but also by time $t$. For instance, a semi-discretization of a parameter-dependent heat equation yields

$$M_h(\boldsymbol{\mu})\, \partial_t U_h(t;\boldsymbol{\mu}) = -A_h(\boldsymbol{\mu})\, U_h(t;\boldsymbol{\mu}).$$

Projecting onto the reduced space $V_N = \mathrm{span}(W)$ leads to the reduced dynamical system

$$M_N(\boldsymbol{\mu})\, \partial_t U_N(t;\boldsymbol{\mu}) = -A_N(\boldsymbol{\mu})\, U_N(t;\boldsymbol{\mu}), \qquad M_N(\boldsymbol{\mu}) = W^\top M_h(\boldsymbol{\mu}) W, \quad A_N(\boldsymbol{\mu}) = W^\top A_h(\boldsymbol{\mu}) W.$$

In this setting, the reduced basis is typically constructed from snapshots $U_h(\boldsymbol{\mu}_i, t_j)$ sampling both the parameter space and the time interval.

If the solution exhibits only limited variation over time, a common simplification is to construct the snapshot matrix using only the final-time solutions $U_h(\boldsymbol{\mu}_i, T)$, which can be sufficient to capture the dominant parametric dependence while significantly reducing the offline cost.

**(ii) Energy and weighted norms: changing the notion of error** In many applications, accuracy is not measured in the Euclidean norm but in a physically meaningful energy norm induced by the inner product of $V$. Let $X_h \in \mathbb{R}^{N_h \times N_h}$ be the SPD matrix with entries $(X_h)_{ij} = (\phi_i, \phi_j)_V$. Minimizing the energy-norm reconstruction error

$$\sum_{j=1}^{n} \|U_h(\boldsymbol{\mu}_j) - WW^\top X_h U_h(\boldsymbol{\mu}_j)\|_{X_h}^2$$

is equivalent to applying POD to the weighted snapshot matrix $X_h^{1/2} S$. This yields a reduced basis that is optimal with respect to the chosen energy norm.

**(iii) Discrete vs. continuous parameter distributions** Imagine we want to find a low-dimensional basis that minimizes the projection error *over a continuous distribution of parameters* $\boldsymbol{\mu} \in \mathcal{P}$:

$$\mathcal{E}_G(B) = \int_{\mathcal{P}} \|U_h(\boldsymbol{\mu}) - BB^\top U_h(\boldsymbol{\mu})\|_2^2 \, d\boldsymbol{\mu}, \qquad B_G^* = \arg \min_{B^\top B = I_N} \mathcal{E}_G(B).$$

The optimal basis $B_G^*$ consists of the orthonormal eigenvectors of the covariance operator

$$C = \int_{\mathcal{P}} U_h(\boldsymbol{\mu}) U_h(\boldsymbol{\mu})^\top \, d\boldsymbol{\mu} \in \mathbb{R}^{N_h \times N_h}.$$

In practice, we only have a finite set of snapshots $\{U_h(\boldsymbol{\mu}_j)\}_{j=1}^n$, and we minimize the *training error*

$$\mathcal{E}_T(B) = \sum_{j=1}^{n} \|U_h(\boldsymbol{\mu}_j) - BB^\top U_h(\boldsymbol{\mu}_j)\|_2^2, \qquad B_T^* = \arg \min_{B^\top B = I_N} \mathcal{E}_T(B).$$

The discrete approximation of the continuous optimal basis is then obtained by performing a numerical SVD of the snapshot matrix $S = [U_h(\boldsymbol{\mu}_1) \mid \cdots \mid U_h(\boldsymbol{\mu}_n)]$, which provides an accurate surrogate for the leading eigenvectors of $C$. This leads to the decomposition of the error into

$$\mathcal{E}_G(B) \leq \underbrace{\sum_{j=N+1}^{r} \sigma_j^2}_{\textit{rank-}N\textit{ approximation error}} + \underbrace{2 \sup_{(B')^\top B' = I_N} |\mathcal{E}_G(B') - \mathcal{E}_T(B')|}_{\textit{generalization gap}} + \underbrace{\mathcal{E}_T(B) - \mathcal{E}_T(B_T^*)}_{\textit{numerical SVD error}},$$

where $\sigma_j^2$ are the eigenvalues of $C$. To reduce the generalization gap, one can employ weighted quadrature rules, corresponding to a rescaled snapshot matrix $SD^{1/2}$ with $D = \mathrm{diag}(w_1, \ldots, w_n)$.

**(iv) Uniform accuracy: greedy alternatives to POD** While POD minimizes an *average* error, some applications require uniform control over the entire parameter domain. Greedy reduced basis methods address this by directly targeting the worst-case error, for example by minimizing

$$\sup_{\boldsymbol{\mu} \in \mathcal{P}} \|U_h(\boldsymbol{\mu}) - WW^\top U_h(\boldsymbol{\mu})\|_{X_h}.$$

These methods iteratively enrich the reduced space using parameter values at which the current approximation error is maximal, leading to different optimality properties than POD-based constructions.

**Takeaways**

- Reduced basis methods approximate the solution manifold by a low-dimensional subspace.
- POD/SVD yields the optimal basis for average snapshot reconstruction.
- Extensions modify *what is optimized*: time, norm, parameter measure, or worst-case error.
- The choice of reduced basis must reflect the intended accuracy criterion and usage regime.

### 3.4 Theoretical results

We now provide a theoretical framework that explains why reduced models based on low-dimensional linear spaces are effective for the class of parametric elliptic problems considered so far. The key ingredients are regularity of the solution map with respect to the parameters, compactness of the associated solution manifold, and its approximability by low-dimensional subspaces.

**Continuity and compactness** Let $u(\boldsymbol{\mu})$ denote the exact solution of the variational problem introduced in Section 2. We define the solution map

$$u : \mathcal{P} \to V, \qquad \boldsymbol{\mu} \mapsto u(\boldsymbol{\mu}),$$

and the associated solution manifold $\mathcal{M} = \{u(\boldsymbol{\mu}) \in V : \boldsymbol{\mu} \in \mathcal{P}\} = u(\mathcal{P}) \subset V$.

**Theorem 3.9** (Lipschitz continuity of the solution map)**.** *Assume that $a(\cdot, \cdot; \boldsymbol{\mu})$ and $f(\cdot; \boldsymbol{\mu})$ are Lipschitz continuous with respect to $\boldsymbol{\mu}$, uniformly with respect to their inputs. Then the solution map $\boldsymbol{\mu} \mapsto u(\boldsymbol{\mu})$ is Lipschitz continuous from $\mathcal{P}$ to $V$.*

Proof

Let $u = u(\boldsymbol{\mu})$ and $u' = u(\boldsymbol{\mu}')$. Then

$$a(u, v; \boldsymbol{\mu}) = f(v; \boldsymbol{\mu}), \qquad a(u', v; \boldsymbol{\mu}') = f(v; \boldsymbol{\mu}') \qquad \forall v \in V.$$

Subtracting the two equations, and adding and subtracting $a(u', v; \boldsymbol{\mu})$, we obtain

$$a(u - u', v; \boldsymbol{\mu}) = f(v; \boldsymbol{\mu}) - f(v; \boldsymbol{\mu}') + a(u', v; \boldsymbol{\mu}') - a(u', v; \boldsymbol{\mu}).$$

Using the Lipschitz continuity of $a$ and $f$, this yields

$$|a(u - u', v; \boldsymbol{\mu})| \leq (L_f + L_a \|u'\|_V)\, \|v\|_V\, \|\boldsymbol{\mu} - \boldsymbol{\mu}'\|_2.$$

Choosing $v = u - u'$ and invoking the coercivity of $a(\cdot, \cdot; \boldsymbol{\mu})$ together with the stability estimate in Eq. (2.2), the result follows.

**Corollary 3.10** (Compactness of the solution manifold)**.** *The solution manifold $\mathcal{M}$ is compact in $V$.*

Proof

Since $\mathcal{P}$ is compact and the solution map $u$ is continuous, the image $\mathcal{M} = u(\mathcal{P})$ is compact in $V$.

**Analyticity** Compactness alone guarantees the existence of low-dimensional approximations, but does not provide information on their convergence rate. Stronger regularity of the solution map leads to quantitative approximation results. To illustrate this, we consider the one-dimensional elliptic problem

$$-(k(\cdot; \mu) u_x(\cdot; \mu))_x = f(\cdot; \mu) \quad \text{in } (-1, 1),$$

with homogeneous Dirichlet boundary conditions and parameter $\mu \in \mathcal{P} \subset \mathbb{R}$. We set $V = H_0^1(-1, 1)$ and assume that $k(\cdot; \mu) \in L^\infty(-1, 1)$ is uniformly positive and that $f(\cdot; \mu) \in L^2(-1, 1)$ for all $\mu \in \mathcal{P}$.

**Theorem 3.11** (Analyticity of the solution map)**.** *Assume that for each $\mu \in \mathcal{P}$ there exists $0 < \gamma < +\infty$ such that*

$$\left\| \frac{1}{k(\cdot; \mu)} \frac{\partial^m k(\cdot; \mu)}{\partial \mu^m} \right\|_{L^\infty(-1,1)} \leq \gamma^m m!, \qquad \frac{1}{1 + \|f(\cdot; \mu)\|_{L^2(-1,1)}} \left\| \frac{\partial^m f(\cdot; \mu)}{\partial \mu^m} \right\|_{L^2(-1,1)} \leq \gamma^m m!,$$

*for all $m \in \mathbb{N}$. Then the solution map $\mu \mapsto u(\mu)$ admits an analytic extension to the complex region*

$$\Sigma(\mathcal{P}) = \{z \in \mathbb{C} : |z - \mu| < 2\gamma \;\; \forall \mu \in \mathcal{P}\}.$$

Let $u_h(\cdot;\mu) \in V_h$ denote a finite element approximation of the solution and define the reduced space

$$\tilde{V}_N := \operatorname{span}(u_h(\cdot;\mu_1), \dots, u_h(\cdot;\mu_N)),$$

where $\{\mu_i\}_{i=1}^N$ are the zeros of the Chebyshev polynomial of the first kind. Let $\{\ell_i\}_{i=1}^N$ denote the associated Lagrange polynomials and define

$$\tilde{u}_N(\cdot;\mu) := \sum_{i=1}^N u_h(\cdot;\mu_i)\,\ell_i(\mu) \in \tilde{V}_N.$$

**Theorem 3.12** (Exponential convergence of parametric interpolation)**.** *There exist constants $\eta > 0$ and $C > 0$, independent of $h$ and $N$, such that*

$$\|u_h - \tilde{u}_N\|_{L^\infty(\mathcal{P};V)} := \sup_{\mu\in\mathcal{P}} \|u_h(\cdot;\mu) - \tilde{u}_N(\cdot;\mu)\|_V \le Ce^{-\eta N}.$$

**Kolmogorov $N$-width** The preceding results motivate a more intrinsic notion of approximability that is independent of any particular reduced space construction.

**Definition 3.13** (Kolmogorov $N$-width)**.** The Kolmogorov $N$-width of $\mathcal{M}$ in $V$ is defined by

$$d_N(\mathcal{M};V) := \inf_{V_N\subset V} \sup_{u\in\mathcal{M}} \inf_{u_N\in V_N} \|u - u_N\|_V.$$

In words, the Kolmogorov $N$-width measures how well the set $\mathcal{M}$ can be approximated by $N$-dimensional subspaces of $V$ in the worst case. It represents the smallest possible approximation error that can be achieved by choosing the *best* $N$-dimensional subspace, where the error is measured as the maximum distance between elements of $\mathcal{M}$ and their closest representatives in the subspace. Thus, the Kolmogorov $N$-width quantifies the intrinsic compressibility of $\mathcal{M}$ in the norm of $V$, independently of any particular reduced model construction.

In practice, we consider the discrete manifold

$$\mathcal{M}_h = \{u_h(\cdot;\mu) : \mu \in \mathcal{P}\} \subset V_h,$$

and the associated width

$$d_N(\mathcal{M}_h;V_h) = \inf_{V_N\subset V_h} \sup_{\mu\in\mathcal{P}} \inf_{u_N(\cdot;\mu)\in V_N} \|u_h(\cdot;\mu) - u_N(\cdot;\mu)\|_V.$$

Choosing the specific space $\tilde{V}_N$ defined above yields

$$d_N(\mathcal{M}_h;V_h) \le \sup_{\mu\in\mathcal{P}} \|u_h(\cdot;\mu) - \tilde{u}_N(\cdot;\mu)\|_V \le Ce^{-\eta N}.$$

This estimate shows that the Kolmogorov $N$-width decays exponentially for smooth elliptic problems. As a consequence, the singular values associated with snapshot matrices exhibit exponential decay. In particular,

$$\mathcal{E}_G(B_G^*) = \sum_{j=N+1}^r \sigma_j^2 \le |\mathcal{P}|^{1/2}\, d_N(\mathcal{M}_h;V_h) \le C|\mathcal{P}|^{1/2} e^{-\eta N},$$

thereby providing a rigorous justification for reduced basis methods based on SVD or POD constructions.

**Takeaways**

- The solution map $\boldsymbol{\mu} \mapsto u(\boldsymbol{\mu})$ is Lipschitz continuous, hence the solution manifold $\mathcal{M}$ is compact.
- Analytic dependence of $k$ and $f$ on $\mu$ implies analytic dependence of the solution and exponential approximability of $\mathcal{M}$.
- As a result, reduced basis spaces built from few well-chosen snapshots converge exponentially.
- This exponential approximability appears as rapid singular value decay, justifying POD/SVD-based reduced models for elliptic problems.

# 4 Discussion and outlook

The previous sections show that RB methods can be extremely effective in many-query regimes. This picture is particularly favorable for *elliptic* problems, where regularity with respect to parameters often implies rapid convergence of linear reduced spaces.

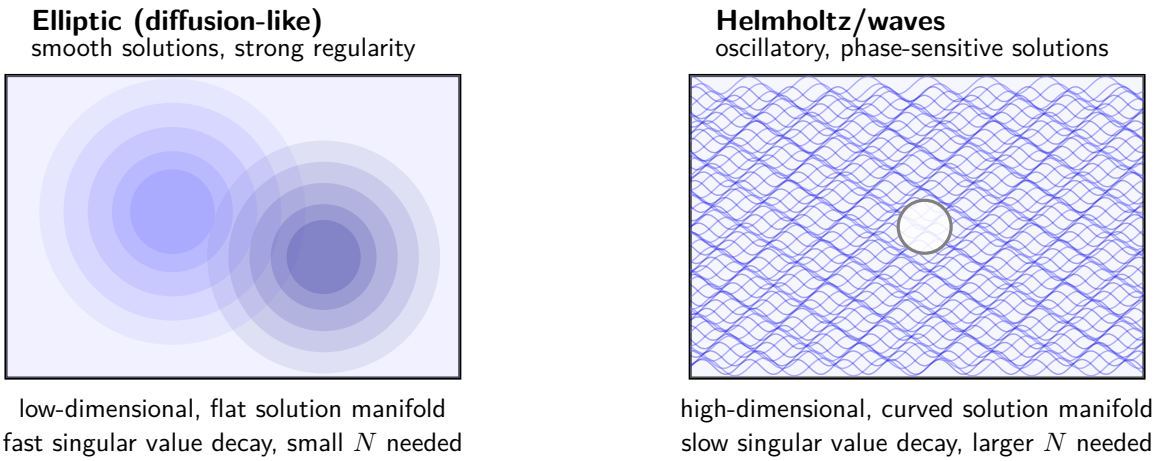



Figure 6: Schematic contrast: elliptic problems often yield smooth, highly compressible solution manifolds, whereas high-frequency wave problems produce oscillatory, phase-sensitive manifolds that are harder to approximate with a single low-dimensional linear space.

**Elliptic problems: the favorable regime** For coercive elliptic operators with smooth (often analytic) parameter dependence, the solution map $\mu \mapsto u(\mu)$ is regular and the solution manifold $\mathcal{M} = u(\mathcal{P})$ is compact and highly approximable. Typical properties:

- Rapid decay of Kolmogorov $N$-widths (often exponential);
- Fast singular value decay in snapshot matrices;
- Accurate reduced spaces with small $N$ via POD/SVD or greedy RB,
- Efficient offline/online separation with stability and speed.

**Wave problems: the challenging regime** For high-frequency wave propagation (e.g., Helmholtz, Maxwell), the solution map loses strong regularity as the wavenumber grows. Typical consequences:

- Kolmogorov $N$-widths decay slowly (often algebraic),
- Singular values decay slowly;
- POD/SVD needs larger $N$;
- Stability is delicate (indefiniteness, resonances, parameter sensitivity).

Thus, a single global linear RB may offer little compression at high frequency, and offline cost can approach the full-order solve.

**Implications for ROM design** This contrast motivates strategies beyond a single global linear space:

- *Local* or *partitioned* bases (parameter clustering, local RB);
- *Phase-aware / physics-informed* ansätze capturing oscillations explicitly;
- *Nonlinear* reduced models (manifold methods, learned embeddings);
- Hybrid RB and operator-learning or data-driven surrogates.

These approaches aim to recover efficient compression when the solution manifold is poorly approximated by a low-dimensional linear space.

# 5 Exercises

**Exercise 5.1** (Reduced Galerkin projection)**.** Let $V_h \subset V$ be a finite-dimensional space and $V_N \subset V_h$ a reduced space with $\dim(V_N) = N$. For a fixed parameter $\mu \in \mathcal{P}$, let $u_h(\mu) \in V_h$ solve

$$a(u_h(\mu), v_h; \mu) = f(v_h; \mu) \qquad \forall v_h \in V_h,$$

and let $u_N(\mu) \in V_N$ solve the reduced Galerkin problem

$$a(u_N(\mu), v_N; \mu) = f(v_N; \mu) \qquad \forall v_N \in V_N.$$

(a) Show that the reduced solution $u_N(\mu)$ satisfies the orthogonality condition

$$a(u_h(\mu) - u_N(\mu), v_N; \mu) = 0 \qquad \forall v_N \in V_N.$$

(b) Using coercivity and continuity of $a(\cdot, \cdot; \mu)$, show that

$$\|u_h(\mu) - u_N(\mu)\|_V \leq \frac{C(\mu)}{\alpha(\mu)} \inf_{v_N \in V_N} \|u_h(\mu) - v_N\|_V.$$

**Exercise 5.2** (POD and optimal low-rank approximation)**.** Let $S = [U_h(\mu_1)| \cdots |U_h(\mu_n)] \in \mathbb{R}^{N_h \times n}$ be a snapshot matrix, and let $S = U\Sigma V^\top$ be its singular value decomposition with singular values $\sigma_1 \geq \cdots \geq \sigma_r > 0$.

(a) Show that for any matrix $B \in \mathbb{R}^{N_h \times N}$ with $B^\top B = I_N$,

$$\sum_{j=1}^{n} \|U_h(\mu_j) - BB^\top U_h(\mu_j)\|_2^2 = \|S - BB^\top S\|_F^2.$$

(b) Show that the choice $B = [U_1| \cdots |U_N]$ minimizes this quantity over all such $B$.

(c) Deduce that the minimum error is given by

$$\sum_{j=N+1}^{r} \sigma_j^2.$$

**Exercise 5.3** (Kolmogorov $N$-width and ROM accuracy)**.** Let $\mathcal{M}_h = \{u_h(\cdot; \mu) : \mu \in \mathcal{P}\} \subset V_h$ be the discrete solution manifold.

(a) Show that for any reduced space $V_N \subset V_h$,

$$\sup_{\mu \in \mathcal{P}} \|u_h(\cdot; \mu) - P_{V_N} u_h(\cdot; \mu)\|_V \geq d_N(\mathcal{M}_h; V_h),$$

where $P_{V_N}$ denotes the orthogonal projection onto $V_N$.

(b) Using the reduced space $\tilde{V}_N$ constructed from snapshots, show that

$$d_N(\mathcal{M}_h; V_h) \leq \sup_{\mu \in \mathcal{P}} \|u_h(\cdot; \mu) - \tilde{u}_N(\cdot; \mu)\|_V.$$

(c) Explain why exponential decay of $d_N(\mathcal{M}_h; V_h)$ implies rapid decay of POD singular values.

## 6 Suggested reading

Table 1: *TL;DR table of 15 favorite references.*

| | Books | Articles |
|---|---|---|
| Greedy algorithms | [1, 7, 12] | [2, 3, 8, 10, 11] |
| RB for time-dependent problems | [1, 7, 12] | [6] |
| Nonlinear RB methods and NNs | – | [4, 5, 9, 13] |
| Quasimatrices & SVD | [15] | [14] |

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